# USHODAYA HIGH SCHOOL

Class X Social Studies • FA-1 Project Work

## INDIA: MAJOR RELIEF FEATURES




*Illustration of India's varied landforms. (Use a textbook map for exact locations).*

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## 1. Introduction

Relief features are the physical features of the Earth's surface, such as mountains, plains, plateaus, deserts, and coastal areas. India has a wide variety of relief features that make its geography highly diverse. These landforms profoundly influence the climate, river systems, agricultural practices, transport networks, settlement patterns, and the daily lives of the people residing in those regions.

## 2. Objectives of the Project

- To identify the six major relief divisions of India.
- To understand their important physical features, formation, and uses.
- To recognise how different landforms influence people's lives and the economy of the country.

## 3. Major Relief Divisions of India

India's geographical area is broadly divided into six major physical divisions. Each division has distinct geological origins and characteristics.

### A. The Himalayas

The Himalayas are young, fold mountains extending along the northern borders of India. They run in a west-to-east direction, forming an arc that covers a distance of about 2,400 kilometres. They consist of three parallel ranges in their longitudinal extent:

- **The Greater Himalayas (Himadri):** The northernmost range containing the highest peaks (like Mt. Everest and Kanchenjunga) and is covered with snow all year round, acting as the source for perennial rivers.
- **The Lesser Himalayas (Himachal):** Located south of Himadri, known for beautiful hill stations (like Shimla, Mussoorie) and deep valleys like Kashmir and Kullu.
- **The Shivaliks:** The outermost range, made of unconsolidated sediments. The longitudinal valleys between Himachal and Shivaliks are called 'Duns' (e.g., Dehradun).

The Himalayas play a crucial role in influencing India's climate by preventing the cold winter winds from Central Asia from entering the subcontinent and by causing the monsoon winds to shed their moisture.

### B. The Indo-Gangetic Plain

This extensive, flat plain was formed over millions of years by the alluvial deposits brought down by three major river systems: the Indus, the Ganga, and the Brahmaputra, along with their tributaries. Spreading over 7 lakh square kilometres, it is divided into three sections:

- **The Western Part:** Formed by the Indus and its tributaries (mostly in Pakistan and Punjab).
- **The Central Part (Ganga Plain):** Extending from Haryana to West Bengal, drained by the Ganga and its tributaries like Yamuna, Ghaghara, and Kosi.
- **The Eastern Part:** Primarily the Brahmaputra valley in Assam.

Based on variations in relief, the plains are divided into Bhabar (porous, pebbly belt), Terai (marshy, heavily forested region), Bhangar (older alluvium), and Khadar (newer, highly fertile alluvium). The rich soil and abundant water make this region the agricultural heartland of India.

### C. The Peninsular Plateau

The Peninsular Plateau is a tableland and is the oldest landmass of India, formed by the breaking and drifting of the Gondwana land. It is composed of old crystalline, igneous, and metamorphic rocks. The plateau is broadly divided into two main parts:

- **The Central Highlands:** Lying to the north of the Narmada river, including the Malwa plateau, bordered by the Vindhya range and the Aravallis.
- **The Deccan Plateau:** A triangular landmass south of the Narmada. It is bounded by the Western Ghats on the west and the Eastern Ghats on the east. The Western Ghats are continuous and higher, while the Eastern Ghats are discontinuous and eroded by rivers.

## D. The Thar Desert

The Thar Desert, also known as the Great Indian Desert, lies towards the western margins of the Aravalli Hills, mainly in western Rajasthan. It is an undulating sandy plain covered with sand dunes (especially crescent-shaped dunes called barchans). The region experiences an arid climate with extremely low rainfall, typically less than 150 mm per year.

Due to the harsh climate, vegetation cover is very sparse. Streams appear only during the short rainy season and soon disappear into the sand, lacking enough water to reach the sea. Luni is the only large river in this region. The Indira Gandhi Canal has recently brought significant changes, allowing for the cultivation of crops in parts of the desert and altering traditional settlement patterns.

## E. The Coastal Plains

The Peninsular Plateau is flanked by narrow coastal strips running along the Arabian Sea on the west and the Bay of Bengal on the east. They play a vital role in India's domestic and international maritime trade, fishing, and agriculture (like coconut and rice farming).

| Feature | Western Coastal Plains | Eastern Coastal Plains |
|---|---|---|
| **Location** | Between Western Ghats and Arabian Sea. | Between Eastern Ghats and Bay of Bengal. |
| **Width & Rivers** | Narrow plains. Rivers form estuaries (no deltas). | Wider and level. Major rivers form large, fertile deltas. |
| **Sub-divisions** | Konkan (North), Kannad (Central), Malabar (South). | Northern Circars (North), Coromandel Coast (South). |

## F. The Islands

India has two major island groups that offer distinct ecosystems, high biodiversity, and are strategically important for coastal and maritime security.

- **Lakshadweep Islands:** Located in the Arabian Sea, close to the Malabar coast of Kerala. This group is composed entirely of small coral islands. They cover a small area (32 sq km) and are known for their rich flora and fauna. Kavaratti island is the administrative headquarters.
- **Andaman and Nicobar Islands:** An elongated chain of islands extending from north to south in the Bay of Bengal. They are larger in size and are more numerous and scattered. They are believed to be an elevated portion of submarine mountains, with some islands being of volcanic origin (like Barren Island, India's only active volcano). They experience an equatorial climate and have thick forest cover.

## 4. Importance of Relief Features

The diverse relief features of India do not exist in isolation; they complement each other and make the country richer in natural resources. Their importance can be summarized as follows:

- **Climate Control:** The Himalayas act as a massive barrier, trapping monsoon winds to bring rain to the subcontinent and blocking freezing winds from the north.
- **Water Security:** Snow-capped mountain peaks are the perennial source for major rivers like the Ganga and Indus, providing water for millions.
- **Food Production:** The vast Indo-Gangetic Plains and coastal deltas possess fertile alluvial soils, making them the "granaries of the country" and supporting dense populations.
- **Industrial Growth:** The Peninsular Plateau is a storehouse of minerals, providing the necessary raw materials for India's heavy industries and manufacturing sector.
- **Economic Activities:** Coastal regions and islands support fishing, port activities, and tourism, deeply integrating India into global trade networks.

## 5. What I Learned

Through this project, I learned that India is a land of great physical contrasts, comprising six major relief divisions: the Himalayas, the Indo-Gangetic Plain, the Peninsular Plateau, the Thar Desert, the Coastal Plains, and the Islands. I understood that each landform has its own unique physical characteristics, climate conditions, and natural resources, all of which heavily dictate the lifestyle, occupations, and economic prosperity of the people living there.

## 6. Conclusion

India's varied relief features make the country geographically and culturally diverse. The interaction between these landforms creates a unique environment that shapes the nation. Understanding these geographical divisions is essential for appreciating the differences in agriculture, resource distribution, and the diverse ways of life found across the country.

> **Project Activity: Map Work**
> Please paste a labelled physical map of India from your textbook or atlas here. Ensure that you have accurately outlined and marked the six major relief divisions discussed in this project using appropriate colors and a legend.

### Bibliography

Telangana State Class X Social Studies Textbook, Chapter 1: "India: Relief Features".
NCERT Class IX Contemporary India-I (Reference for geographical expansion).

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